

BADIN, Pakistan

WMO: 417850

Lat: 24.633N

Lon: 68.900E

Elev: 36

StdP: 14.68

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	48.9	51.3	30.9	25.1	70.7	34.6	29.4	68.2	12.0	72.2	10.5	71.3	0.7	50	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	16.5	105.6	76.9	103.3	76.9	101.3	77.3	83.3	95.1	82.6	94.2	81.9	93.0	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.8	88.2	80.0	156.2	87.6	79.4	152.8	87.2	47.5	96.0	46.7	94.2	46.0	93.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.2	12.4	DB	45.2	110.4	5.7	2.0	41.1	111.8	37.8	113.0	34.6	114.1	30.5	115.5	(4)
(5)			WB	42.2	85.5	3.4	1.2	39.8	86.3	37.9	87.0	36.0	87.7	33.5	88.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.3	64.9	70.9	80.1	87.5	91.5	91.6	88.6	86.0	84.9	76.1	67.2	(6)	
(7)		DBStd	9.62	4.07	5.47	4.64	3.00	2.19	2.26	2.48	1.95	2.07	2.57	4.50	4.47	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	93	54	13	0	0	0	0	0	0	0	0	0	26	(9)
(10)		CDD50	11431	462	584	933	1125	1285	1249	1196	1117	1080	1082	784	534	(10)
(11)		CDD65	6050	51	177	468	675	820	799	731	652	630	617	335	95	(11)
(12)		CDH74	78543	461	1587	5129	8947	12014	12045	10390	8449	8141	7546	3088	746	(12)
(13)		CDH80	43230	54	542	2746	5405	7596	7731	5943	4033	3927	3851	1254	148	(13)
(14)	Wind	WSAvg	4.8	2.0	2.5	3.6	5.6	8.9	8.6	8.0	6.4	5.2	2.7	1.6	1.9	(14)
(15)	Precipitation	PrecAvg	9.7	0.1	0.3	0.0	0.1	0.2	0.4	2.8	3.3	2.1	0.4	0.1	0.0	(15)
(16)		PrecMax	35.5	0.6	2.4	0.2	0.7	6.3	2.5	11.9	14.0	13.8	4.6	1.4	0.3	(16)
(17)		PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	7.6	0.1	0.6	0.0	0.2	1.2	0.6	3.3	3.9	3.7	1.1	0.3	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.1	93.5	102.7	107.3	109.1	108.2	102.1	96.5	99.1	99.9	94.0	86.7	(19)
(20)			MCWB	63.1	67.3	71.3	73.8	76.4	79.5	81.8	82.4	80.9	77.5	71.3	66.9	(20)
(21)		2%	DB	81.4	89.9	98.9	104.3	106.1	104.0	98.8	94.5	96.3	97.2	91.5	83.9	(21)
(22)			MCWB	61.9	66.7	70.9	74.0	77.3	80.6	81.3	80.9	81.0	76.1	70.9	65.1	(22)
(23)		5%	DB	79.0	86.4	95.7	102.2	103.7	101.4	96.8	92.7	93.9	95.3	89.6	81.2	(23)
(24)			MCWB	61.4	65.6	69.3	73.5	77.4	80.7	81.1	80.3	80.2	75.7	70.5	63.9	(24)
(25)		10%	DB	76.3	83.1	92.8	99.6	101.4	99.3	94.9	91.3	92.2	93.7	87.3	78.2	(25)
(26)			MCWB	59.5	63.7	68.4	73.2	77.8	80.8	80.7	80.0	79.5	75.6	69.7	61.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.3	72.1	76.0	79.7	83.3	84.1	84.2	83.8	83.9	81.6	76.7	71.1	(27)
(28)			MCD B	79.9	86.3	93.9	98.5	99.0	99.2	95.6	93.3	93.9	93.9	87.5	82.3	(28)
(29)		2%	WB	65.6	70.1	73.7	78.1	81.9	82.8	82.8	82.7	82.8	80.1	74.7	68.3	(29)
(30)			MCD B	76.5	83.7	91.8	95.7	96.7	96.8	93.7	91.1	92.1	91.3	85.2	78.8	(30)
(31)		5%	WB	63.3	68.4	72.3	76.9	81.1	82.3	82.1	81.7	81.8	79.2	73.2	66.2	(31)
(32)			MCD B	74.6	80.6	89.1	92.6	94.9	95.7	92.6	89.7	90.7	90.2	83.9	77.3	(32)
(33)		10%	WB	61.1	66.6	70.9	76.0	80.3	81.7	81.5	80.9	80.9	78.3	71.8	64.2	(33)
(34)			MCD B	73.5	79.1	86.4	91.4	93.3	94.5	91.6	88.6	89.2	88.8	82.7	75.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.7	23.6	24.9	25.2	21.8	16.5	12.4	10.8	13.0	18.3	21.3	21.5	(35)
(36)			MCDBR	25.3	26.5	28.7	28.3	25.4	20.2	15.4	13.0	16.1	20.4	22.2	23.6	(36)
(37)			MCWBR	9.7	7.6	6.5	6.4	5.0	3.8	3.1	3.5	4.6	4.7	6.3	8.7	(37)
(38)		5% WB	MCD BR	22.3	23.0	25.7	25.8	20.9	17.0	13.7	12.1	13.9	16.9	19.2	20.7	(38)
(39)			MCWBR	11.0	8.0	7.3	6.7	4.4	3.6	3.8	4.4	4.8	4.5	6.6	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.443	0.469	0.530	0.584	0.642	0.724	0.778	0.709	0.610	0.558	0.539	0.487	(40)	
(41)		taud	2.020	1.987	1.847	1.767	1.697	1.587	1.534	1.648	1.810	1.826	1.818	1.909	(41)	
(42)		Ebn at Noon	252	256	247	237	222	204	193	206	224	227	220	231	(42)	
(43)		Edh at Noon	49	54	65	72	77	86	90	81	67	63	60	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1363	1641	1934	2155	2227	1987	1605	1585	1761	1722	1411	1280	(44)	
(45)		RadStd	48	52	62	63	54	123	113	91	92	66	56	58	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page